

Recent Trends in Statistics and Data Science

by

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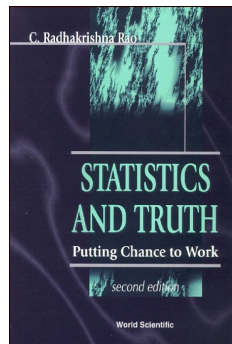
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- **Big data** is an all-encompassing term for any collection of data sets so large and complex that it becomes difficult to process using traditional data processing applications.
- **Artificial Intelligence** research is defined as the study of intelligent agents: any device that perceives its environment and takes actions that maximize its chance of success at some goal.
- **Machine learning** is the sub-field of computer science that gives computers the ability to learn without being explicitly programmed.
- **Data science** is the study of the generalizable extraction of knowledge from data, yet the key word is science.
- **Statistics** is the study of the collection, analysis, interpretation, presentation and organization of data.

- All knowledge is, in final analysis, history.
All sciences are, in the abstract, mathematics.
All judgements are, in their rationale, statistics.
 - Professor C R Rao, Professor Emeritus at Pennsylvania State University, Jan 1987.
- When you're fundraising, it's AI.
When you're hiring, it's ML.
When you're implementing, it's Linear Regression.
When you're debugging, it's *printf()*.
 - Baron Schwartz, Founder and CEO of VividCortex, Nov 2017.



- **Statistics** is the universal tool of inductive inference, research in natural and social sciences, and technological applications.

Statistics, therefore, must always have purpose, either in the pursuit of knowledge or in the promotion of human welfare.

- Professor P C Mahalanobis.

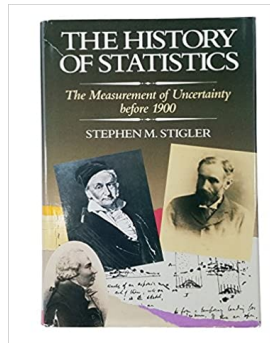
- **Role of Statistics:**

- ① making inference from samples
- ② development of new methods for complex data sets
- ③ quantification of uncertainty and variability

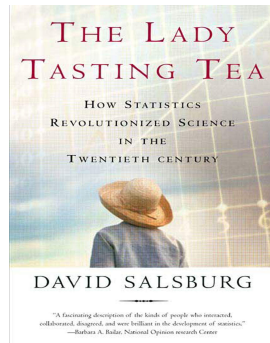
- **Two Views of Statistics:**

- ① Statistics as a Mathematical Science
- ② Statistics as a Data Science

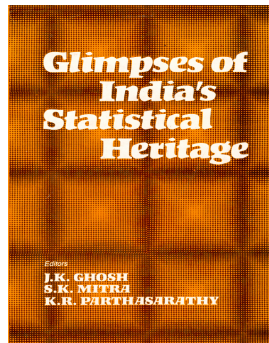
- Probability and its application to Gambling & Astronomy
 - Jacob Bernoulli (Prob. distribution, Law of large numbers, etc.)
 - Pierre-Simon Laplace (Double exponential, Transformation, etc.)
 - Thomas Bayes (Bayes' theorem, etc.)
 - Gauss and Legendre (Least Square Method)
 - Francis Galton (Correlation & Regression)
 - Karl Pearson (χ^2 test, distribution, etc.)
 - and many others.



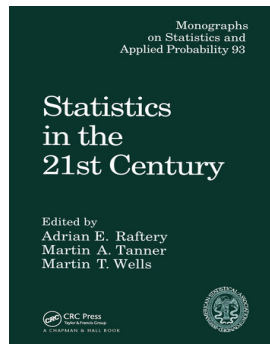
- Development of Statistics and its application to Agriculture, Economics, Geology, Medical, Technology, Clinical Trials, etc.
 - Ronald Fisher (Discriminant analysis, Likelihood, ANOVA & DOE, etc.)
 - Jerzy Neyman, Egon Sharpe Pearson & Wald (Decision theory, Optimality, etc.)
 - Lehmann, Hotelling, Anderson & Tukey (Multivariate & Inferential Statistics, etc.)
 - Box, Cox, Jenkin and Blackwell (Time Series)
 - Shewhart, Deming, Taguchi & Juran (SQC)
 - Efron, Breiman, Friedman, Cramer (Modern Statistical Tools)
 - and many others.



- In a speech at ISI, Professor R A Fisher once pointed out that more than half the qualified statisticians working in the world were Indians, for quite some time. Most of them were Professor Rao's students.
 - (Professor) PC Mahalanobis (Mahalanobis Distance, Founder of ISI.)
 - C. R. Rao (Linear Models, Multivariate Analysis, Orthogonal arrays, etc.)
 - J K Ghosh (Bayesian & High-dimensional Statistics, etc.)
 - Samarendra Nath Roy (Multivariate Statistics)
 - Raj Chandra Bose (Combinatorial Design)
 - Debabrata Basu (Statistical Inference)
 - and many others.



- Parametric Models : One Sample, two sample, linear models, survival data, Estimation, Testing of Hypothesis.
- Probability distributions were believed to generate data (e.g., Gaussian, Logistic, Poisson, Exponential, etc.).
- Semiparametric & Nonparametric Models : Dropping assumptions on population, dependence and errors.
- Emphases on Optimality in various ways : Bayes optimality, Decision theory, minimax and unbiasedness.
- Exact distributional (t, F) approaches and asymptotic methods (samples size $\rightarrow \infty$ viewed as approximation).



Few Famous Quotations : Beyond Normality

- “Normality is a myth; there never was, and there never will be a normal distribution”
 - Roy C. Geary (1947; Biometrika, vol. 34, 248).
- “Everybody believes in the exponential law of errors (i.e. the normal distribution), the experimenters, because they think it can be proved by mathematicians; and the mathematicians, because they believe that it has been established by observations”
 - E.T. Whittaker and G. Robinson (1967).
- “... the statisticians knows ... that in nature there never was a normal distribution, there never was a straight line, yet with normal and linear assumptions, known to be false he can often derive results which match to a useful approximation, those found in real world”
 - George W. Box (1976, Jour. Amer. Stat. Asso., vol. 71, 791-799).

- Data : Large bodies of data with complex data structures are generated from computers, sensors, manufacturing industries, etc.
- Models : Non/Semiparametric models but in complex probability spaces / high-dimensional functional spaces (e.g., deep neural net, reinforcement learning, decision trees, etc.).
- Emphases : Making predictions, causation, algorithmic convergence.
- **Data** are necessary and at the core of Statistical Learning, Data Science & Machine Learning.

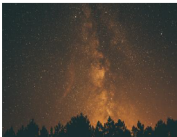
- **Probability** : Has moved to the center of Mathematics and having strong interactions with Statistical Physics and Theoretical Computer Science.
- **Statistics** : Not only has strong interactions with Probability but also other parts of Data Science (Machine Learning, Artificial Intelligence, etc.).
- **Computational** : Computing skills are essential, construction of fast training algorithms and computation time.
- **Applications** : Strong interactions with substantive fields in all areas. Applications of statistical methods in almost all the fields are evident. Statistics became a key technology driven by data (**“Data is the new oil”**).

2018 *This Is What Happens In An Internet Minute*



The World is Data Rich

Astronomy



Social Networks



Healthcare



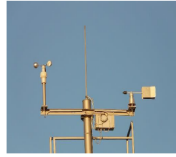
Banking



Genomics

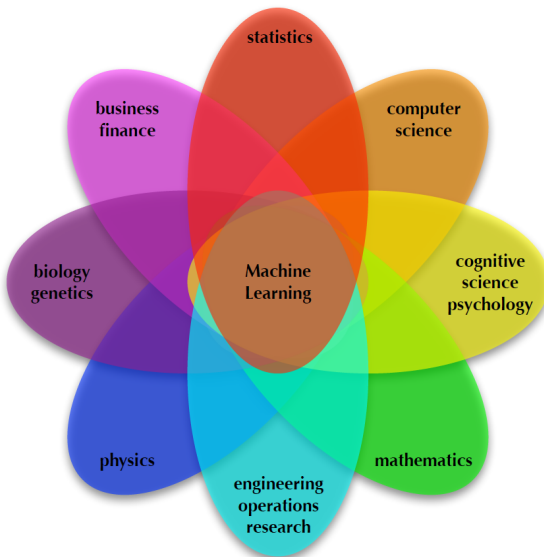


Weather measurements



What is Machine Learning?

Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.



What is the Difference?

- Traditional Problems in Applied Statistics:

- Well formulated question that we would like to answer.
- Expensive to gathering data and/or expensive to do computation.
- Create specially designed experiments to collect high quality data.

- Current Situation : Information Revolution

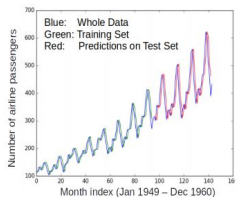
- Improvements in computers and data storage devices.
- Powerful data capturing devices.
- Lots of data with potentially valuable information available.

What is the Difference?

- Data characteristics:
 - Size
 - Dimensionality
 - Complexity
 - Messy
 - Secondary sources
- Focus on generalization performance :
 - Prediction on new data
 - Action in new circumstances
 - Complex models needed for good generalization
- Computational considerations :
 - Large scale and complex systems

Introduction to Machine Learning

- Designing algorithms that **ingest data** and **learn a model** of the data.
- The learned model can be used to
 - ① Detect **patterns/structures/themes/trends** etc. in the data
 - ② Make **predictions** about future data and make decisions



- Modern ML algorithms are heavily **“data-driven”**.
- Optimize a performance criterion using example data or **past experience**.

- **Unsupervised Learning:**

- Uncover structure hidden in 'unlabelled' data.
- Given network of social interactions, find communities.
- Given shopping habits for people using loyalty cards: find groups of 'similar' shoppers.
- Given expression measurements of 1000s of genes for 1000s of patients, find groups of functionally similar genes.
- Goal: Hypothesis generation, visualization.

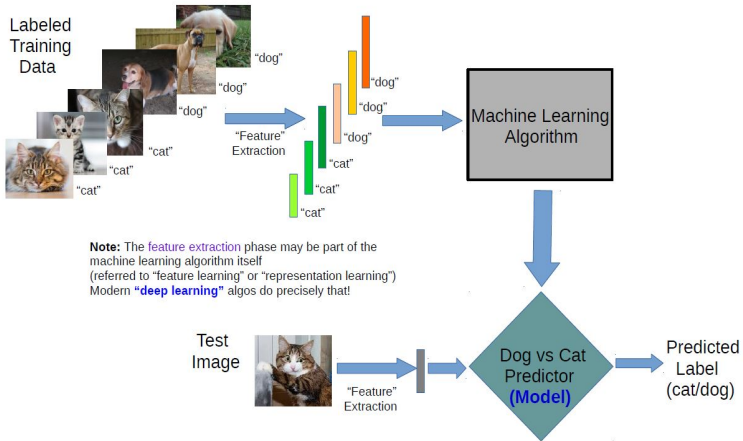
- **Supervised Learning:**

- A database of examples along with 'labels' (task-specific).
- Given expression measurements of 1000s of genes for 1000s of patients along with an indicator of absence or presence of a specific cancer, predict if the cancer is present for a new patient.
- Given network of social interactions along with their browsing habits, predict what news might users find interesting.
- Goal: Prediction on new examples.

- **Semi-supervised Learning:**
 - A database of examples, only a small subset of which are labelled.
- **Multi-task Learning:**
 - A database of examples, each of which has multiple labels corresponding to different prediction tasks.
- **Reinforcement Learning:**
 - An agent acting in an environment, given rewards for performing appropriate actions, learns to maximize its reward.

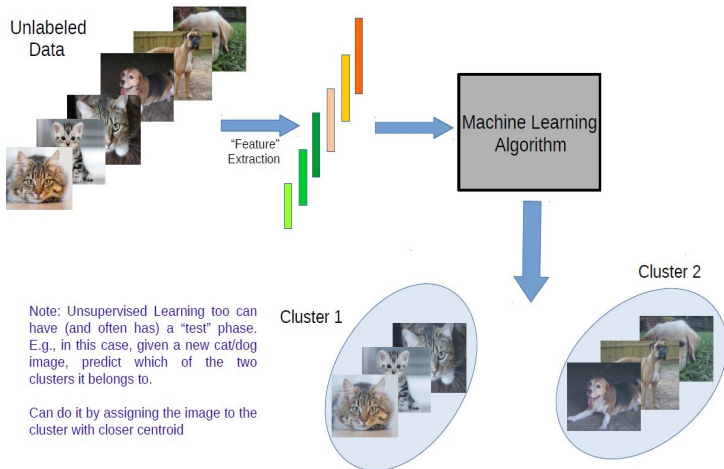
A Typical Supervised Learning Workflow (for Classification)

Supervised Learning: Predicting patterns in the data



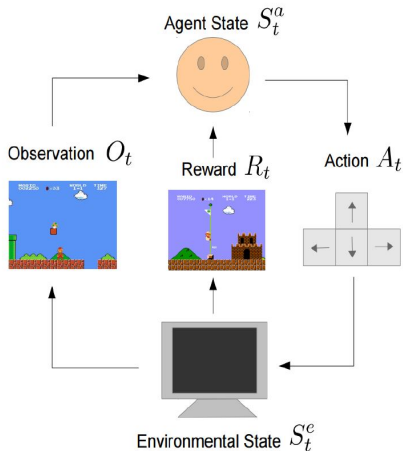
A Typical Unsupervised Learning Workflow (for Clustering)

Unsupervised Learning: Discovering patterns in the data



A Typical Reinforcement Learning Workflow

Reinforcement Learning: Learning a "policy" by performing actions and getting rewards (e.g, robot controls, beating games)



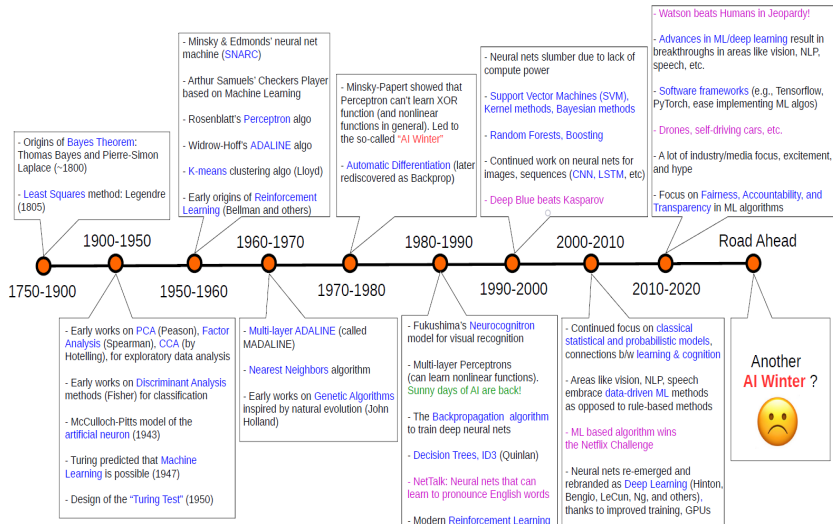
Agent's goal is to learn a policy for some task

Agent does the following repeatedly

- Senses/observes the environment
- Takes an action based on its current policy
- Receives a reward for that action
- Updates its policy

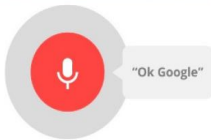
There IS supervision, not explicit (as in Supervised Learning) but rather implicit (feedback based)

Parallel Progress in Statistics & Machine Learning



Machine Learning in the real-world

Broadly applicable in many domains (e.g., internet, robotics, healthcare and biology, computer vision, NLP, databases, computer systems, finance, etc.).



Predictive Policing



Online Fraud Detection

Machine Learning helps Natural Language Processing

ML algorithm can learn to translate text

English ▾



Welcome to this
course Edit

Hindi ▾



इस कोर्स में आपका स्वागत है
is kors mein aapaka svaagat hai



(even "transliterate")

Machine Learning meets Speech Processing

ML algorithms can learn to translate speech in real time

PUTTING MACHINE LEARNING TO THE TEST
To provide a seamless user experience, Skype Translator uses machine learning to solve key challenges in interpreting human language, including:



Representing the different ways people really speak



Determining sentence boundaries, punctuation and ease from speech

there
they're
their

Disambiguating sound-alike words in context



Mapping words and phrases from one language to another

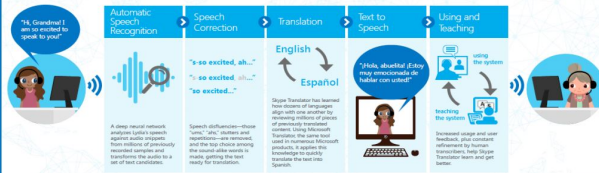


NOW YOU'RE SPEAKING MY LANGUAGE (LITERALLY)



Skype has always been about making it easy to talk with family and friends all over the world. Now, by integrating advanced speech recognition and automatic translation into Skype, Skype Translator lets you speak with those you've always wished you could, even if they speak a different language.

HOW SKYPE TRANSLATOR WORKS



TRANSLATE INSTANT MESSAGES IN OVER 40 LANGUAGES

Having a translated IM conversation is super easy! Choose a contact, turn on the Translation switch for that person, and start typing. When you're ready to "tap send", your original message will appear in the right-hand pane, followed by its translation. Your contact on the other end will see something very similar, albeit with the translated message in his/her preferred language presented first. While voice translation initially supports English and Spanish only, IM translation supports over 40 languages, so feel free to experiment with them all...even Klingon!



Register for the preview at www.skype.com/translator and wait for your invite.

Install the Skype Translator client.

Use Skype Translator to call someone who speaks Spanish. Or, if you speak Spanish, call someone who speaks English.

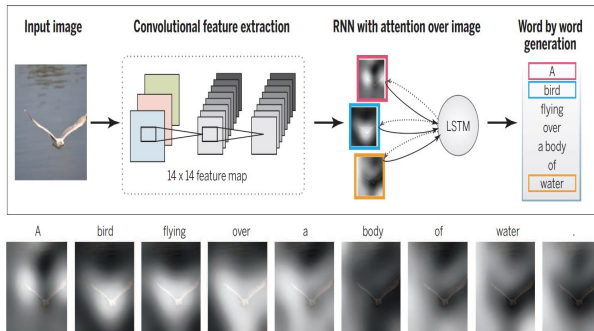
Every call you make helps Skype Translator get a little bit better. You won't see the improvement right away, but you will see gradual improvement over time.

Machine Learning helps Computer Vision

- Automatic generation of text captions for images:

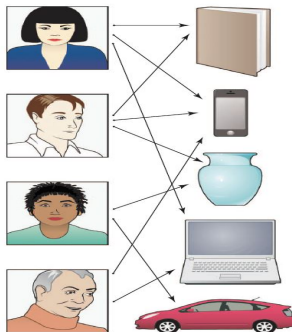
A **convolutional neural network** is trained to interpret images, and its output is then used by a recurrent neural network trained to generate a text caption.

- The sequence at the bottom shows the word-by-word focus of the network on different parts of input image while it generates the caption word-by-word.



Machine Learning helps Recommendation systems

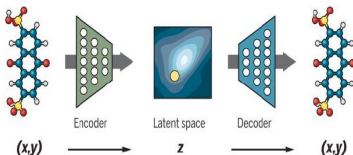
- A **recommendation system** is a machine-learning system that is based on data that indicate links between a set of users (e.g., people) and a set of items (e.g., products).
- A link between a user and a product means that the user has indicated an interest in the product in some fashion (perhaps by purchasing that item in the past).
- The **machine-learning problem** is to suggest other items to a given user that he or she may also be interested in, based on the data across all users.



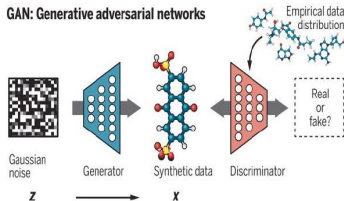
Machine Learning helps Chemistry

ML algorithms can understand properties of molecules and learn to synthesize new molecules¹.

VAE: Variational autoencoders

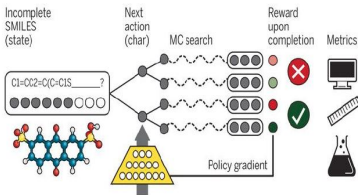


GAN: Generative adversarial networks

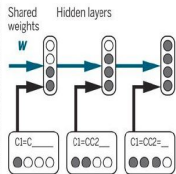


RL: Reinforcement learning

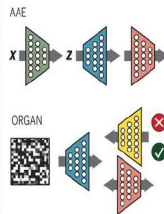
Policy gradient with Monte Carlo tree search (MCTS)



RNN: Recurrent neural network



Hybrid approaches



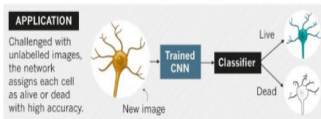
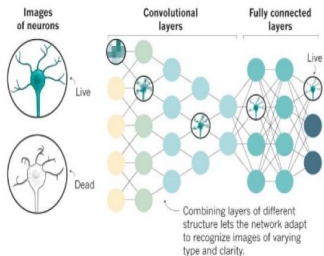
¹Inverse molecular design using machine learning: Generative models for matter engineering (Science, 2018)

Machine Learning helps Image Recognition



Machine Learning helps Many Other Areas...

Biology



©nature

Finance



- Emerging Research Areas in Statistics, Data Science & ML:
 - All the areas of Machine Learning.
 - High-Dimensional Small Sample Size Data.
 - Spatio-temporal Data.
 - Fairness in ML and Interpretable Deep Learning.
 - Collaborative research with other disciplines.
- Statistical Thinking:
 - Accepting randomness in formulating models and ideas.
 - Realizing and Analyzing dependence of conclusions on assumptions.
 - Measuring uncertainty in some ways without forgetting dependence on assumptions.

Start with these three books



Then you can start reading these books

